Runs on every example: figures by run and model

These runs follow `docs/paper_plan.md`. They were made on 2026-10-11 with three decision models: Jev, Liquid d1 and GPT-6 Luna Decisions. The conclusions come first. Each finding then has a page with a chart, evidence, one example and a table for the paper. The terms follow, then one table per run with the models as rows. `python scripts/report/runs_report.py paper_runs --out docs/paper_runs.html` rebuilds this document from the results folders and `scripts/definitions/paper_runs.json`. Claude Code read the figures on 2026-10-11 and wrote the conclusions and their strength. Rushab set the claim and the strength of finding 2 on 2026-10-11. Rushab has not reviewed the others.

Conclusions

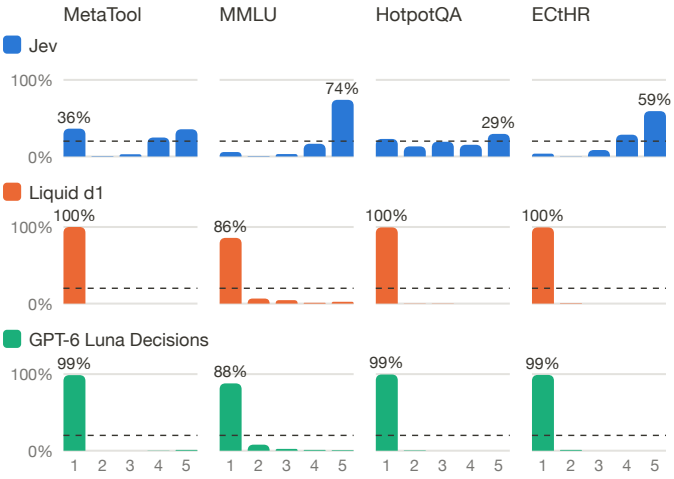
1. The claim. A decision model gives each option in a list a probability. That probability does not reliably say whether the option is correct. It is close to the accuracy when exactly one option in the list is correct. It is not close in the three other cases.
2. Several correct options (finding 2, the main finding). When 2 to 4 options in a list are correct, one of them has a mean probability of 0.66 to 0.85. The correct option with the lowest probability has 0.11 to 0.19. This happens on three datasets and three models. TypeSafe's example code notifies a second team when its probability is above 0.25. Every correct option is above 0.25 in 15% to 35% of the answers. Strong.
3. TypeSafe's example (finding 6). TypeSafe's example ticket has two issues. Jev gives their two teams 0.65 and 0.31. The documentation shows 0.61 and 0.35. On 24 tickets written the same way, both teams are above 0.25 in 11 tickets on Jev, 12 on Liquid d1 and 3 on GPT-6 Luna Decisions. When an issue is sent alone, the models give its team 0.99 to 1.00. Moderate: 24 tickets.
4. Why the second correct option is low (finding 5). Remove the correct option with the highest probability from the list. The second correct option then rises from between 0.11 and 0.19 to between 0.53 and 0.71. Sharing out the removed probability in proportion predicts this rise. Liquid d1 returns full-precision probabilities: the prediction is 0.64, 0.80 and 0.70 on the three datasets, and the model gives 0.61, 0.70 and 0.67. Jev and GPT-6 Luna Decisions return 2 decimals, and their values agree within 0.13 on the examples where 0.10 or more is left. This supports finding 2. It is not a finding of its own.
5. Identical options (finding 1). In a list of 5 copies of one correct option, every copy is equally correct, and equal probabilities are 0.20. Jev gives its highest copy 0.29 to 0.34 and its lowest copy 0.10 to 0.12. Liquid d1 gives them 0.50 to 0.81 and 0.03 to 0.08. GPT-6 Luna Decisions gives them 0.67 to 0.79 and 0.01 to 0.02. Liquid d1 and GPT-6 Luna Decisions pick the first copy in 86% to 100% of their answers. Strong.
6. No correct option (finding 3). When no option in a list is correct, Jev and Liquid d1 answer every list. Their mean confidence is 0.49 to 0.87. GPT-6 Luna Decisions refuses 58% to 98% of the lists with random options. It refuses 1% to 27% of the lists with the example's distractors. Strong for random options.
7. One correct option (finding 4). The mean confidence is within 0.14 of the accuracy. The probability works in this case. When several options are correct, the mean confidence is 0.08 to 0.23 below the accuracy.
8. For the paper. Finding 2 carries the paper. Finding 6 tests it on TypeSafe's own example. Findings 1 and 3 show two more cases where the probability does not say whether an option is correct. Each request was sent once. No figure has a margin of error yet.

1. Identical options: the copies do not get equal probabilities, and the position decides which copy gets the most

A list of 5 copies of one correct option has no better option, and equal probabilities are 0.20 each. No model gives equal probabilities. Jev gives its highest copy 0.29 to 0.34 and its lowest copy 0.10 to 0.12. Liquid d1 gives them 0.50 to 0.81 and 0.03 to 0.08, and GPT-6 Luna Decisions 0.67 to 0.79 and 0.01 to 0.02. The position decides which copy is highest.

Strength for the paper: strong for the three models. Identical options get different probabilities: the highest copy is ahead of the lowest copy by 0.17 to 0.23 on Jev, 0.42 to 0.78 on Liquid d1 and 0.65 to 0.78 on GPT-6 Luna Decisions. Liquid d1 and GPT-6 Luna Decisions put the highest copy first. Jev's preferred position changes with the dataset.

Share of predictions at each position for 5 copies of one correct option



Evidence

- Jev gives the highest copy a mean of 0.32, 0.33, 0.34 and 0.29 on MetaTool, MMLU, HotpotQA and ECtHR, and the lowest copy 0.11, 0.10, 0.11 and 0.12. The highest copy gets 2.4 to 3.3 times the probability of the lowest copy.
- Liquid d1 gives the highest copy 0.81, 0.50, 0.73 and 0.55, and the lowest copy 0.03, 0.08, 0.04 and 0.07. GPT-6 Luna Decisions gives the highest copy 0.78, 0.74, 0.79 and 0.67, and the lowest copy 0.02, 0.01, 0.01 and 0.02.
- The highest copy is ahead of the second-highest copy by 0.06 to 0.11 on Jev, 0.31 to 0.73 on Liquid d1 and 0.45 to 0.68 on GPT-6 Luna Decisions.
- The mean entropy is 1.03 to 1.85 bits on Liquid d1, 1.00 to 1.35 bits on GPT-6 Luna Decisions and 2.18 to 2.24 bits on Jev. Equal probabilities give 2.32 bits.
- Liquid d1 picks position 1 in 99.9%, 85.7%, 99.5% and 99.6% of answers on MetaTool, MMLU, HotpotQA and ECtHR, respectively. GPT-6 Luna Decisions picks it in 98.6%, 87.8%, 99.4% and 98.9% of answers, respectively. An equal share is 20%.
- Jev picks the last position in 73.9% of answers on MMLU, 59.3% on ECtHR, 35.5% on MetaTool and 29.5% on HotpotQA.
- With 5 paraphrases of one MetaTool tool, Liquid d1 picks position 1 in 74% of answers. GPT-6 Luna Decisions picks it in 87% of answers, and Jev in 26%.
- With one correct tool among distractors, GPT-6 Luna Decisions has an accuracy of 0.80 when the tool is first and 0.67 when it is last. Jev's accuracy goes from 0.78 to 0.80, and Liquid d1's from 0.69 to 0.74.
- Each model received 30,695 lists across four datasets. Each list appeared in its 5 cyclic permutations.

One example

Of the 1,000 MMLU questions, this question has probabilities by position closest to the three models' means.

Query. Which of the following groups or organizations can be identified as having the characteristics of a bureaucracy?

Instruction. Choose the correct answer to the question.

The list sent to the model: 5 copies of the correct choice

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**The government of a large city**	0.12	**0.48**	**0.74**
2	**The government of a large city**	0.10	0.13	0.16
3	**The government of a large city**	0.19	0.11	0.04
4	**The government of a large city**	0.26	0.11	0.04
5	**The government of a large city**	**0.33**	0.17	0.02

`mmlu/test/13562` in `results/mmlu_copied/2026-10-11_01`

Limits

- The copies' names differ by extra spaces. The API needs those spaces to tell them apart.
- A user does not send a list of identical options. The run with one correct tool among distractors connects this test to real lists. There, the effect is 0.13 for one model and 0.05 or less for the two others.
- Each request was sent once. Jev does not always give the same answer to the same request.

Table for the paper

Each list has 5 copies of one correct option and is sent in its 5 cyclic permutations. Every copy is a correct option. The table shows the mean probability of the highest copy and of the lowest copy, and the share of predictions at the first position. Equal probabilities are 0.20, and an equal share is 20%.

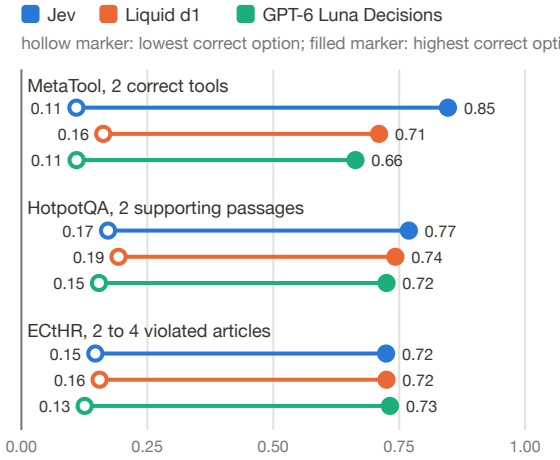
	MetaTool			HotpotQA			ECtHR		
Model	Highest correct option	Lowest correct option	Predictions at the first position	Highest correct option	Lowest correct option	Predictions at the first position	Highest correct option	Lowest correct option	Predictions at the first position
Jev	0.32	0.11	36.2%	0.34	0.11	22.9%	0.29	0.12	3.6%
Liquid d1	0.81	0.03	99.9%	0.73	0.04	99.5%	0.55	0.07	99.6%
GPT-6 Luna Decisions	0.78	0.02	98.6%	0.79	0.01	99.4%	0.67	0.02	98.9%

2. Allocation: the model gives most of the probability to one of several correct options

When a list has 2 to 4 correct options, the models give most of the probability to one of them: a mean of 0.66 to 0.85. The lowest correct option gets 0.11 to 0.19. The highest distractor gets 0.04 to 0.17, and the two ranges overlap. Asking for both options does not change this. On Jev, "Choose both." gives the lowest correct MetaTool tool 0.15, and "Choose one of them." gives it 0.14.

Strength for the paper: strong, and the paper's main finding. TypeSafe's documentation reads 0.61 and 0.35 as a ticket for two teams. Its example code notifies a second team above 0.25. Across three datasets and three models, every correct option is above 0.25 in 15% to 35% of answers. Someone who follows that rule misses a correct option in 65% to 85% of these lists.

Mean probability for the lowest and highest correct options



Evidence

- On MetaTool, the mean probabilities for the highest and lowest correct options are 0.85 and 0.11 for Jev, 0.71 and 0.16 for Liquid d1, and 0.66 and 0.11 for GPT-6 Luna Decisions. On HotpotQA, the same models give 0.77 and 0.17, 0.74 and 0.19, and 0.72 and 0.15. On ECtHR, they give 0.72 and 0.15, 0.72 and 0.16, and 0.73 and 0.13.
- On MetaTool, every correct option is above 0.25 in 15% of Jev's answers, 27% of Liquid d1's answers, and 16% of GPT-6 Luna Decisions' answers. In the same model order, this happens in 30%, 35% and 26% of HotpotQA answers, and in 23%, 25% and 17% of ECtHR answers.
- TypeSafe's primer says outcomes with a probability of `0.2` should happen about 20% of the time. The models give the lowest correct option a mean probability of 0.11 to 0.19. The dataset labels it correct in every one of these lists.
- On average, the lowest correct option is 0.05 to 0.14 above the highest distractor. For GPT-6 Luna Decisions on MetaTool, it is 0.06 below. A probability threshold does not separate the second correct option from the distractors.
- The models usually rank the options correctly. The prediction is a correct option in 0.82 to 0.98 of answers. The 2 highest options are the 2 correct ones in 0.52 to 0.85 of answers.
- With Jev on MetaTool, "Choose one of them." gives the lowest correct tool 0.14, and "Choose both." gives it 0.15. On HotpotQA, "may support", "supports" and "definitely supports" give the lowest supporting passage 0.18, 0.17 and 0.17.
- Finding 5 explains the cause. Finding 6 tests TypeSafe's example. When an issue is sent alone, the model gives its team 0.99 to 1.00. When the ticket has a second issue, the team with the lower probability gets 0.12 to 0.28 on average.
- The results cover 7,405 HotpotQA questions, 497 MetaTool queries and 192 ECtHR cases.

One example

This HotpotQA question is closest to the three models' means among questions with 10 passages where the three models rank the same supporting passage first.

Query. Who is younger, Tony Roche or Renáta Tomanová?

Instruction. Choose every passage in the list that supports the answer to the question.

The 10 passages of the question

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**1976 Marlboro NSW Open** The 1976 New South Wales Open, also known by its sponsored name Marlboro New South W…	0.00	0.00	0.07
2	**Pavel Vojtisek** Paul Vojtischek (born Pavel Vojtíšek, 13 June 1963) is a Czechoslovak-born West Germ…	0.01	0.01	0.01
3	**1976 Australian Open – Women's Singles** Evonne Goolagong defeated Renáta Tomanová 6–2, 6–2 in the final to win the Women's S…	0.00	0.00	0.07
4	**Renáta Tomanová** correct Renáta Tomanová (born 9 December 1954) is a former professional tennis player from C…	**0.81**	**0.79**	**0.76**
5	**1980 French Open – Mixed Doubles** The Mixed Doubles tournament at the 1980 French Open was held from 26 May to 8 June…	0.00	0.00	0.00
6	**1979 Australian Open – Women's Doubles** Betsy Nagelsen and Renáta Tomanová were the defending champions.	0.00	0.00	0.01
7	**Tony Roche** correct Anthony Dalton Roche, AO MBE (born 17 May 1945) is a former professional Australian…	0.18	0.20	0.08
8	**1974 Australian Indoor Championships – Doubles** Rod Laver and John Newcombe were the defending champions but only Newcombe competed…	0.00	0.00	0.00
9	**Olga Zaitseva (tennis)** Olga Zaitseva (born 12 March 1962) is a former competitor for the Soviet Union at th…	0.00	0.00	0.00
10	**1976 French Open – Women's Singles** Two time reigning champion Chris Evert did not defend her title. She would stay home…	0.00	0.00	0.00

`hotpotqa/validation/5a7dcdea5542990b8f503abc` in `results/hotpotqa_removal/2026-10-11_01`

Limits

- TypeSafe's documentation gives the 0.25 threshold. It does not say what probabilities two correct options should get. The finding is that this threshold misses correct options. It does not say that 0.50 each is the right answer.
- HotpotQA labels a passage when it has a supporting sentence. The passage may be needed to answer the question even if it does not support the answer on its own, as finding 5's yes/no questions show.
- The ECtHR labels show the Court's findings. They do not cover every article related to the case facts.

Table for the paper

For lists with several correct options, the table shows the mean probability of the highest correct option, the lowest correct option and the highest distractor. It also shows the share of answers in which every correct option is above 0.25, the threshold in TypeSafe's example code.

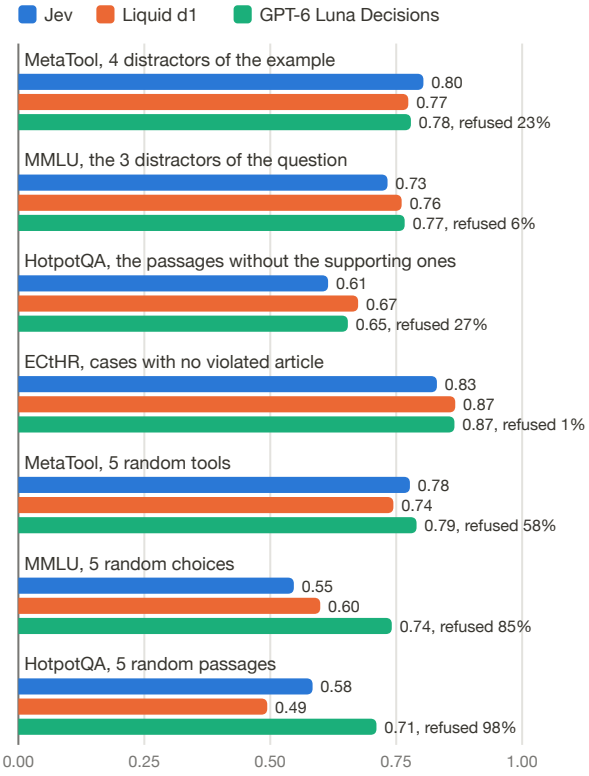
	MetaTool, 2 correct tools				HotpotQA, 2 supporting passages				ECtHR, 2 to 4 violated articles			
Model	Highest correct option	Lowest correct option	Highest distractor	Answers with every correct option above 0.25	Highest correct option	Lowest correct option	Highest distractor	Answers with every correct option above 0.25	Highest correct option	Lowest correct option	Highest distractor	Answers with every correct option above 0.25
Jev	0.85	0.11	0.04	15%	0.77	0.17	0.05	30%	0.72	0.15	0.07	23%
Liquid d1	0.71	0.16	0.12	27%	0.74	0.19	0.05	35%	0.72	0.16	0.07	25%
GPT-6 Luna Decisions	0.66	0.11	0.17	16%	0.72	0.15	0.08	26%	0.73	0.13	0.06	17%

3. Abstention: confident answers to lists with no correct option

Jev and Liquid d1 answer every list with no correct option. Their confidence is close to their confidence on lists with one correct option. GPT-6 Luna Decisions returns an answer of type `refusal` for many lists with no correct option. On the lists it answers, its confidence is as high as Jev's and Liquid d1's. When NOTA is offered, the three models pick it for random options. They pick it less often for the example's distractors.

Strength for the paper: strong for lists of random options, where no option can be correct. Moderate for the example's distractors, where the label says no option is correct. The vendor does not document GPT-6 Luna Decisions' refusals.

Mean confidence on lists with no correct option and no NOTA



Evidence

- On lists of 5 random options, Jev's mean confidence is 0.78, 0.55 and 0.58 on MetaTool, MMLU and HotpotQA. Liquid d1's is 0.74, 0.60 and 0.49 in the same order.
- On four datasets, Jev's mean confidence is 0.61 to 0.83 on lists of the example's distractors. Liquid d1's is 0.67 to 0.87. On lists with one correct option, Jev's is 0.86 to 0.94 and Liquid d1's is 0.81 to 0.89.
- GPT-6 Luna Decisions refuses 58%, 85% and 98% of lists with random options. It refuses 23%, 6%, 27% and 1% of lists with the example's distractors. Jev and Liquid d1 refuse none of 202,088 questions each.
- When lists have NOTA and random options, the models predict NOTA in 0.93 to 1.00 of their answers. With NOTA and the example's distractors, they predict it in 0.54 to 0.72 of answers on MMLU and 0.79 to 0.88 on HotpotQA. For ECtHR cases with no violated article, they predict NOTA in 0.37, 0.07 and 0.17 of answers.
- One request with a refused list was sent again 7 times, with one change each time. The model answered the same passages when the query was changed to one that a passage fits. It answered the same query when NOTA or a fitting passage was added to the list.

One example

Among queries whose first list GPT-6 Luna Decisions refused, this MetaTool query is closest to Jev and Liquid d1's mean confidence.

Query. My child is interested in space, can you suggest any children's picture books about space exploration?

Instruction. Pick all the tools in the list that are relevant to the task at hand.

5 tools outside the query's tool list

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**SSH** Ability to SSH into your server and turn your natural language into server commands.	0.12	**0.66**	refused
2	**WeatherTool** Provide you with the latest weather information.	0.14	0.09	refused
3	**GifApi** Search through a wide range of gifs - Powered by Giphy.	**0.70**	0.21	refused
4	**airqualityforeast** Planning something outdoors? Get the 2-day air quality forecast for any US zip code.	0.02	0.02	refused
5	**haulingbuddies** Locate dependable animal transporters using recommendations, reviews, and regulatory…	0.02	0.02	refused

`metatool/scenario/1797` in `results/metatool_unrelated/2026-10-11_01`

Limits

- A MetaTool example's distractors can include a tool that fits the query. The run's label was read for only 50 queries.
- In an ECtHR case with no label, the Court found no violation. The case's facts can still point to an article.
- GPT-6 Luna Decisions answers 2% to 99% of the lists. Its figures cover only the lists it answers.
- No source says why the model refuses. The API gives no reason.

Table for the paper

Lists with no correct option and no NOTA: the share of the lists that the model refused, and its mean confidence on the lists it answered.

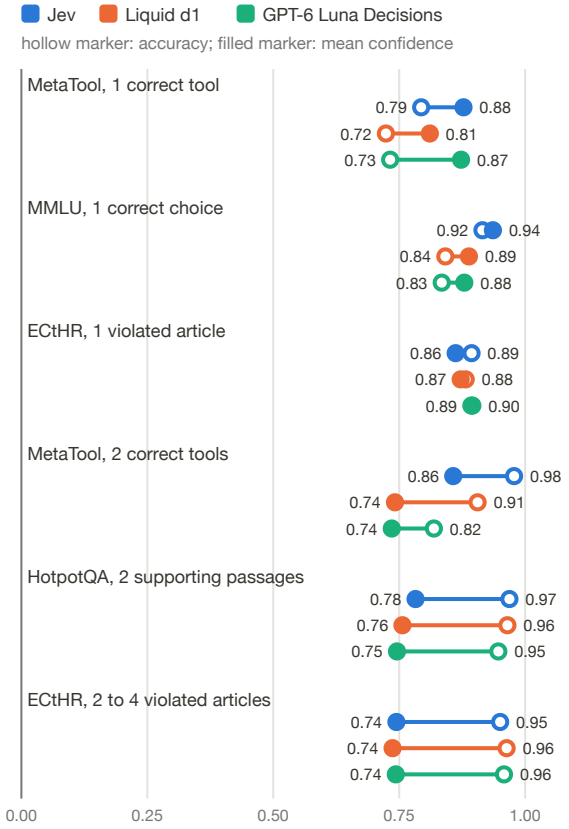
Model	MetaTool, 5 random tools		HotpotQA, 5 random passages		MetaTool, 4 distractors		HotpotQA, no supporting passage		ECtHR, no violated article	
	Lists refused	Mean confidence	Lists refused	Mean confidence	Lists refused	Mean confidence	Lists refused	Mean confidence	Lists refused	Mean confidence
Jev	0%	0.78	0%	0.58	0%	0.80	0%	0.61	0%	0.83
Liquid d1	0%	0.74	0%	0.49	0%	0.77	0%	0.67	0%	0.87
GPT-6 Luna Decisions	58%	0.79	98%	0.71	23%	0.78	27%	0.65	1%	0.87

4. Calibration: confidence is close to accuracy with one correct option, but the result does not carry over

On lists with one correct option, the three models' confidence is close to their accuracy. With several correct options, their confidence is 0.08 to 0.23 below their accuracy. With none, their confidence is 0.49 to 0.87. A confidence threshold set on one kind of list does not hold on another. With several correct options, a yes/no question per option gives each option a probability. On MetaTool, the tools given 0.9 to 1.0 for yes are correct in 80% to 89% of the answers, and those given 0.2 to 0.3 in 24% to 40%.

Strength for the paper: the case in which the probability works, not a finding on its own. The figures are sound. The gaps on the other lists follow from findings 2 and 3. A shared probability lowers confidence, and a list with no correct option has accuracy 0.

Accuracy and mean confidence



Evidence

- With one correct option, mean confidence minus accuracy is +0.08, +0.09 and +0.14 on MetaTool for Jev, Liquid d1 and GPT-6 Luna Decisions. It is +0.02, +0.05 and +0.04 on MMLU, and -0.03, -0.01 and 0.00 on ECtHR.
- With several correct options, mean confidence minus accuracy is -0.12, -0.16 and -0.08 on MetaTool. It is -0.19, -0.21 and -0.20 on HotpotQA, and -0.21, -0.23 and -0.21 on ECtHR.
- With one correct option, ECE with 10 bins ranges from 0.01 to 0.15.
- With a yes/no question per option, the answers from 0.9 to 1.0 for yes, at mean probabilities of yes of 0.93 to 0.99, are about a correct option in 80% to 100% of the cases. The answers from 0.2 to 0.3, at mean probabilities of 0.25 to 0.26, are about a correct option in 49% to 53% on HotpotQA, 24% to 40% on MetaTool and 6% to 35% on ECtHR.
- On the 401 HotpotQA bridge questions, the passages given 0.2 to 0.3 for yes, at mean probabilities of 0.24 to 0.25, are supporting passages in 46% to 51% of the answers, and the passages given 0.9 to 1.0, at 0.94 to 0.98, in 80% to 83%.
- The analysis includes 16,770 MetaTool lists, 4,000 MMLU lists and 6,550 ECtHR lists with one correct option.

One example

This example is among the MetaTool queries where Jev is wrong with a confidence of 0.90 or more on the released order. It has the shortest list, with the lowest id breaking a tie.

Query. Yes, I can definitely assist you with that. Could you please specify whether you prefer a step-by-step explanation or would you like me to provide you with a sample code snippet as well?

Instruction. Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None.

The query's tool list and "None", in the released order

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**RepoTool** correct Discover GitHub projects tailored to your needs, explore their structures with insig…	0.02	0.01	0.00
2	**AI2sql** Converts a natural language text into an SQL query.	0.00	0.00	0.00
3	**SSH** Ability to SSH into your server and turn your natural language into server commands.	0.00	0.00	0.00
4	**AutoInfra1** Talk to your Servers. Works with AWS, GCP, Azure, and anywhere you can ssh!	0.00	0.00	0.00
5	**noteable** Create notebooks in Python, SQL, and Markdown to explore data, visualize, and share…	0.00	0.00	0.00
6	**dart** Project management on autopilot.	0.00	0.00	0.00
7	**hackit_web_scanner** AI Powered Web Scanner by HACKIT.	0.00	0.00	0.00
8	**LarkBaseImporter** Importing data into Lark Base for further analysis and presentation. An easy-to-use…	0.00	0.00	0.00
9	**webhooks** Allows you to write, deploy, and manage HTTP Webhooks in JavaScript, right from the…	0.00	0.00	0.00
10	**universal** Enables to access web pages, analyze PDFs, generate QR codes, etc.	0.00	0.00	0.00
11	**None**	**0.98**	**0.99**	**1.00**

`metatool/scenario/0` in `results/metatool_position/2026-10-11_01`

Limits

- Rushab, 2026-10-14: calibration of a list's probabilities is not defined when several options are correct, since the probabilities of a list sum to 1. With several correct options, only the yes/no questions are used.
- Rushab, 2026-10-15: the yes/no answers go into 10 bins of width 0.1, as for ECE. A bin holds 4 to 3,895 answers, and 16 of the 90 bins of the first table hold fewer than 50: 6 of GPT-6 Luna Decisions on ECtHR, 5 of GPT-6 Luna Decisions on MetaTool, 4 of Liquid d1 on ECtHR and 1 of Jev on ECtHR.
- Accuracy counts a prediction as correct when it is one of the correct options. With several correct options, a different definition gives a different gap.
- MetaTool labels one tool per query. On some queries, the three models agree on a different answer, as in the example.
- No interval has been computed yet.

Table for the paper

Accuracy and mean confidence for lists with one correct option and lists with several correct options. A prediction is correct when it is one of the correct options.

Model	MetaTool, 1 correct tool		ECtHR, 1 violated article		MetaTool, 2 correct tools		HotpotQA, 2 supporting passages		ECtHR, 2 to 4 violated articles	
	Accuracy	Mean confidence	Accuracy	Mean confidence	Accuracy	Mean confidence	Accuracy	Mean confidence	Accuracy	Mean confidence
Jev	0.79	0.88	0.89	0.86	0.98	0.86	0.97	0.78	0.95	0.74
Liquid d1	0.72	0.81	0.88	0.87	0.91	0.74	0.96	0.76	0.96	0.74
GPT-6 Luna Decisions	0.73	0.87	0.90	0.89	0.82	0.74	0.95	0.75	0.96	0.74

Yes/no answers by probability of yes

Yes/no questions about each option of the examples with two or more correct options: the 497 MetaTool queries with 2 correct tools, 500 HotpotQA questions with 2 supporting passages and the 192 ECtHR cases with 2 to 4 violated articles. The answers are put in 10 bins of width 0.1 by probability of yes. A cell gives the share of the answers in the bin that are about a correct option, and the number of answers. A calibrated probability of yes equals that share. Refused questions are left out.

Probability of yes (mean probability)	Jev, MetaTool	Jev, HotpotQA	Jev, ECtHR	Liquid d1, MetaTool	Liquid d1, HotpotQA	Liquid d1, ECtHR	GPT-6 Luna Decisions, MetaTool	GPT-6 Luna Decisions, HotpotQA	GPT-6 Luna Decisions, ECtHR
0.0 to 0.1 (0.00 to 0.06)	1% (3,369)	4% (3,277)	0% (882)	1% (3,356)	7% (3,746)	0% (1,156)	7% (3,895)	7% (3,801)	4% (1,360)
0.1 to 0.2 (0.14 to 0.15)	19% (259)	28% (532)	2% (306)	17% (201)	38% (332)	2% (134)	36% (85)	37% (183)	29% (51)
0.2 to 0.3 (0.25 to 0.26)	40% (119)	52% (289)	6% (113)	24% (141)	53% (196)	13% (70)	38% (55)	49% (124)	35% (20)
0.3 to 0.4 (0.34 to 0.36)	52% (117)	53% (189)	24% (80)	36% (73)	57% (94)	13% (30)	34% (29)	47% (91)	36% (39)
0.4 to 0.5 (0.45 to 0.47)	57% (115)	56% (158)	41% (86)	38% (87)	67% (119)	31% (32)	52% (27)	63% (73)	49% (152)
0.5 to 0.6 (0.55 to 0.57)	62% (131)	67% (106)	57% (106)	49% (57)	63% (63)	47% (36)	66% (32)	59% (56)	94% (36)
0.6 to 0.7 (0.65 to 0.66)	70% (176)	65% (98)	84% (121)	70% (83)	68% (66)	56% (27)	51% (37)	57% (54)	88% (17)
0.7 to 0.8 (0.75 to 0.77)	78% (265)	84% (77)	88% (163)	51% (137)	79% (98)	60% (70)	53% (47)	58% (66)	94% (17)
0.8 to 0.9 (0.84 to 0.86)	68% (288)	83% (83)	95% (59)	69% (226)	77% (102)	78% (90)	61% (69)	73% (102)	91% (32)
0.9 to 1.0 (0.93 to 0.99)	89% (131)	81% (146)	100% (4)	85% (609)	83% (139)	95% (275)	80% (694)	81% (401)	95% (156)

HotpotQA yes/no answers by probability of yes, by type of question

The HotpotQA yes/no answers of the previous table, split by HotpotQA's type of question: 401 bridge questions and 99 comparison questions. A passage of a comparison question holds one of the two entities, so it may not support the answer alone. Fewer than 50 answers fall in 22 of the 30 bins of the comparison questions and in 2 of the 30 bins of the bridge questions.

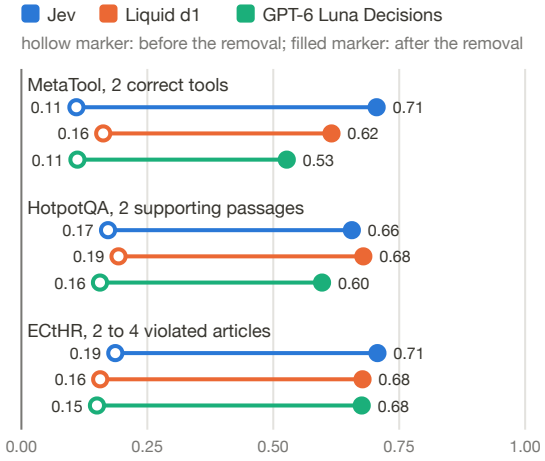
Probability of yes (mean probability)	Jev, bridge	Jev, comparison	Liquid d1, bridge	Liquid d1, comparison	GPT-6 Luna Decisions, bridge	GPT-6 Luna Decisions, comparison
0.0 to 0.1 (0.01 to 0.04)	4% (2,661)	4% (616)	7% (2,967)	11% (779)	7% (3,081)	10% (720)
0.1 to 0.2 (0.14 to 0.15)	27% (395)	33% (137)	33% (251)	53% (81)	35% (107)	39% (76)
0.2 to 0.3 (0.24 to 0.25)	46% (194)	62% (95)	49% (159)	70% (37)	51% (84)	45% (40)
0.3 to 0.4 (0.34 to 0.35)	50% (135)	61% (54)	53% (77)	76% (17)	54% (50)	39% (41)
0.4 to 0.5 (0.45 to 0.46)	55% (120)	58% (38)	69% (99)	60% (20)	60% (42)	68% (31)
0.5 to 0.6 (0.54 to 0.57)	68% (98)	50% (8)	67% (57)	33% (6)	52% (44)	83% (12)
0.6 to 0.7 (0.63 to 0.67)	66% (91)	57% (7)	72% (60)	33% (6)	56% (50)	75% (4)
0.7 to 0.8 (0.75 to 0.80)	84% (76)	100% (1)	79% (91)	71% (7)	54% (59)	86% (7)
0.8 to 0.9 (0.85 to 0.88)	84% (77)	67% (6)	77% (97)	80% (5)	72% (92)	80% (10)
0.9 to 1.0 (0.94 to 0.98)	80% (142)	100% (4)	83% (131)	88% (8)	81% (380)	71% (21)

5. Removal: the second correct option has a low probability because the first has most of it

An option's probability is its share of the list. It does not judge that option alone. When the correct option with the highest probability is removed, the second correct option gets most of that option's probability. Dividing the removed probability among the remaining options in proportion gives nearly the same value.

Strength for the paper: supports finding 2. It is not a separate finding. Probabilities in a list sum to 1, so the remaining options' probabilities rise when one is removed. Liquid d1 returns probabilities with full precision. Dividing its second correct option's probability in the full list by the probability of the remaining options gives 0.64, 0.80 and 0.70 on MetaTool, HotpotQA and ECtHR. After the removal, Liquid d1 gives 0.61, 0.70 and 0.67. Jev and GPT-6 Luna Decisions return 2 decimals, so their division is less exact. Across the three models, the two values are within 0.13 of each other in each of the nine cells.

The second correct option's mean probability before and after removal



Evidence

- On MetaTool, Jev's probability for the second correct option goes from 0.11 to 0.71, Liquid d1's from 0.16 to 0.62, and GPT-6 Luna Decisions' from 0.11 to 0.53 after removal. On HotpotQA, their probabilities go from 0.17 to 0.66, 0.19 to 0.68 and 0.16 to 0.60. On ECtHR, they go from 0.19 to 0.71, 0.16 to 0.68 and 0.15 to 0.68.
- On the examples where the options other than the removed one hold 0.10 or more, the second correct option's renormalised probability is 0.77, 0.64 and 0.40 on MetaTool for Jev, Liquid d1 and GPT-6 Luna Decisions. It is 0.80, 0.80 and 0.61 on HotpotQA, and 0.72, 0.70 and 0.59 on ECtHR. After the removal, the same examples give 0.72, 0.61 and 0.49 on MetaTool, 0.67, 0.70 and 0.60 on HotpotQA, and 0.71, 0.67 and 0.68 on ECtHR.
- When one correct option is removed, the models predict the remaining correct option in 0.86, 0.78 and 0.67 of the MetaTool answers and in 0.84, 0.84 and 0.80 of the HotpotQA answers.
- In a yes/no question about a correct MetaTool tool alone, the models give yes a mean probability of 0.64, 0.79 and 0.65. They give 66% to 85% of those tools 0.50 or more for yes.
- In a yes/no question about a violated ECtHR article alone, the models give yes a mean probability of 0.67, 0.89 and 0.74. They give 80% to 96% of those articles 0.50 or more for yes.
- When asked about a supporting HotpotQA passage alone, the models give it 0.45, 0.40 and 0.52. They give 38% to 51% of those passages 0.50 or more.

One example

The question from finding 2 is shown after the removal.

Query. Who is younger, Tony Roche or Renáta Tomanová?

Instruction. Choose every passage in the list that supports the answer to the question.

The list for the question from finding 2 leaves out Renáta Tomanová, the supporting passage with the highest probability. In the full list, the models give Tony Roche 0.18, 0.20 and 0.08.

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**1976 Marlboro NSW Open** The 1976 New South Wales Open, also known by its sponsored name Marlboro New South W…	0.01	0.07	**0.44**
2	**Pavel Vojtisek** Paul Vojtischek (born Pavel Vojtíšek, 13 June 1963) is a Czechoslovak-born West Germ…	0.34	0.22	0.03
3	**1976 Australian Open – Women's Singles** Evonne Goolagong defeated Renáta Tomanová 6–2, 6–2 in the final to win the Women's S…	0.01	0.05	0.08
4	**1980 French Open – Mixed Doubles** The Mixed Doubles tournament at the 1980 French Open was held from 26 May to 8 June…	0.01	0.03	0.01
5	**1979 Australian Open – Women's Doubles** Betsy Nagelsen and Renáta Tomanová were the defending champions.	0.01	0.01	0.01
6	**Tony Roche** correct Anthony Dalton Roche, AO MBE (born 17 May 1945) is a former professional Australian…	**0.60**	**0.53**	0.41
7	**1974 Australian Indoor Championships – Doubles** Rod Laver and John Newcombe were the defending champions but only Newcombe competed…	0.00	0.00	0.00
8	**Olga Zaitseva (tennis)** Olga Zaitseva (born 12 March 1962) is a former competitor for the Soviet Union at th…	0.01	0.06	0.00
9	**1976 French Open – Women's Singles** Two time reigning champion Chris Evert did not defend her title. She would stay home…	0.01	0.01	0.02

`hotpotqa/validation/5a7dcdea5542990b8f503abc` in `results/hotpotqa_removal/2026-10-11_01`

Limits

- Jev and GPT-6 Luna Decisions return probabilities with 2 decimals. Dividing by a remainder near 0.01 means little at 2 decimals. The renormalised value therefore uses only the examples where the remaining options hold 0.10 or more. On Jev these are 227 of 497 MetaTool queries, 4,866 of 7,384 HotpotQA questions and 150 of 192 ECtHR cases.
- The HotpotQA yes/no figures show that about half of the supporting passages are not judged to support the answer alone. This weakens the view that both passages are correct options.
- HotpotQA's yes/no questions cover 500 questions.

Table for the paper

The first two columns show the second correct option's mean probability in the full list and after removing the correct option with the highest probability. The last two columns use the examples where the other options hold 0.10 or more: its full-list probability divided by the probability of those options, and its probability after the removal. Jev and GPT-6 Luna Decisions return probabilities with 2 decimals.

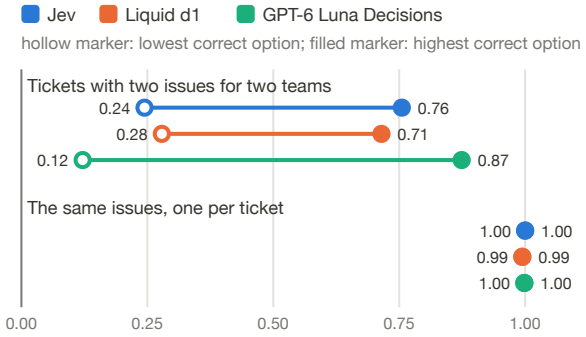
	MetaTool, 2 correct tools				HotpotQA, 2 supporting passages				ECtHR, 2 to 4 violated articles			
Model	Second correct option in the full list	After the removal	Full list, renormalised	After the removal, same examples	Second correct option in the full list	After the removal	Full list, renormalised	After the removal, same examples	Second correct option in the full list	After the removal	Full list, renormalised	After the removal, same examples
Jev	0.11	0.71	0.77	0.72	0.17	0.66	0.80	0.67	0.19	0.71	0.72	0.71
Liquid d1	0.16	0.62	0.64	0.61	0.19	0.68	0.80	0.70	0.16	0.68	0.70	0.67
GPT-6 Luna Decisions	0.11	0.53	0.40	0.49	0.16	0.60	0.61	0.60	0.15	0.68	0.59	0.68

6. The documented example: a ticket for two teams

TypeSafe's Choice documentation shows a ticket about a wrong size and a double charge. It gives returns 0.61 and billing 0.35, and says the ticket belongs to two teams. Its example code notifies a second team when its probability is above 0.25. Jev gives that answer again. Across 24 similar tickets, the rule finds the second team in about half of the tickets on Jev and Liquid d1, and in 3 tickets on GPT-6 Luna Decisions.

Strength for the paper: moderate: 24 tickets. It is the direct test of finding 2 on TypeSafe's documented example. The documented answer holds, so the paper cannot say the example is wrong. The rule in the example code is not reliable. On Jev, a team gets 1.00 when its issue is sent alone, and below 0.25 in 13 of 24 tickets with a second issue.

Mean probability for the lower and higher correct teams



Evidence

- TypeSafe's example request was sent once on 2026-10-11. Jev gives returns 0.65, billing 0.31 and shipping 0.04. The documentation shows returns 0.61, billing 0.35 and shipping 0.04. Liquid d1 gives returns 0.20, billing 0.75 and shipping 0.05. GPT-6 Luna Decisions gives returns 0.49, billing 0.50 and shipping 0.01.
- For 24 tickets with two issues, Jev gives the higher of the two teams a mean probability of 0.76 and the lower one 0.24. Liquid d1 gives them 0.71 and 0.28. GPT-6 Luna Decisions gives them 0.87 and 0.12.
- The second team is above 0.25 in 11 of 24 tickets on Jev, 12 tickets on Liquid d1 and 3 tickets on GPT-6 Luna Decisions.
- The same 24 issues were sent one per ticket. Each issue's team gets a mean probability of 1.00 on Jev, 0.99 on Liquid d1 and 1.00 on GPT-6 Luna Decisions.
- The team neither issue concerns gets 0.00 to 0.01. The models separate it from the two teams that apply, but give those two teams different probabilities.
- Changing the order of the two issues changes the prediction in 1 of 12 pairs on Jev, 4 pairs on Liquid d1 and 3 pairs on GPT-6 Luna Decisions.
- TypeSafe's ticket also mentions a late delivery. Shipping gets 0.04 on Jev, 0.05 on Liquid d1 and 0.01 on GPT-6 Luna Decisions.

One example

The ticket with two issues closest to the three models' means.

Query. The tracking page has shown no update for nine days. The boots are a size too small and I need the next size up.

Instruction. Which team should handle this?

The 3 teams in TypeSafe's example

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**returns** correct Exchanges, wrong or damaged items	**0.68**	**0.65**	**0.86**
2	**shipping** correct Delivery status, delays, lost packages	0.32	0.35	0.14
3	**billing** Charges, invoices, payment problems	0.00	0.00	0.00

`tickets/two_issues/11` in `results/tickets_removal/2026-10-11_01`

Limits

- Claude Opus 5.5 wrote the 24 issues on 2026-10-11. No second reader checked that each issue concerns only one team.
- The two issues in a ticket may not matter equally. No reference gives the right split.
- Each request was sent once. When the team question from TypeSafe's ticket was sent again during the ticket run, Jev gave 0.58 and 0.37.
- This test uses one vendor's example and three teams. The four datasets in findings 1 to 5 support the general claim.

Table for the paper

The table covers 24 tickets with two issues for two teams, and the same 24 issues sent one per ticket. It shows the mean probability for the higher correct team, the lower correct team and the highest other team. It also shows the share of tickets where both correct teams are above 0.25.

Model	24 tickets with two issues				The 24 issues, one per ticket	
	Highest correct option	Lowest correct option	Highest distractor	Answers with every correct option above 0.25	Highest correct option	Highest distractor
Jev	0.76	0.24	0.00	46%	1.00	0.00
Liquid d1	0.71	0.28	0.01	50%	0.99	0.01
GPT-6 Luna Decisions	0.87	0.12	0.00	12%	1.00	0.00

Terms

- **Option**: an entry in the list sent with a query. It is a tool on MetaTool, a choice on MMLU, a passage on HotpotQA, or an article on ECtHR.
- **Correct option**: an option the dataset labels as correct for the query.
- **Distractor**: an option that is not a correct option.
- **NOTA**: an option that says no other option is correct. It is "None" on MetaTool, "None of the above" on MMLU, "None of the passages" on HotpotQA, and "None of these articles" on ECtHR. NOTA is the correct option when the list has no other correct option.
- **Prediction**: the option the model picks in the `choice` field of its answer.
- **Confidence**: the highest probability in an answer. It is not the `confidence` field of the model's answer.
- **Refused list**: a list for which the model returns an answer of type `refusal` and no probabilities.
- **Accuracy**: the share of answers whose prediction is a correct option.
- **Exact match**: for a list with k correct options, where k is 2 or more, the share of answers whose k options with the highest probabilities are the k correct options. A tie for place k does not count as a match.
- **Overconfidence**: mean confidence minus accuracy.
- **ECE (expected calibration error)**: the answers go into 10 bins by confidence, from 0 to 0.1 through 0.9 to 1.0. Each bin has a distance between its mean confidence and its accuracy. ECE averages the 10 distances, weighted by the number of answers in each bin.
- **Prediction minus runner-up**: the highest probability in an answer minus the second-highest probability.
- **Highest correct minus lowest correct**: the highest probability of a correct option minus the lowest probability of a correct option.
- **Highest correct minus highest distractor**: the highest probability of a correct option minus the highest probability of a distractor.
- **Lowest correct minus highest distractor**: the lowest probability of a correct option minus the highest probability of a distractor. It is negative when a distractor has a higher probability than a correct option.
- **Entropy**: the sum of -p log2(p) for every probability p in an answer, in bits. It is 0 when one option has 1.00. Its maximum is log2 of the number of options. The maximum occurs when every option has the same probability, and the column header shows it.
- **Position**: an option's place in the list sent to the model, counting from the first.

Notes

- Jev is `jev-1.13.0` and uses TypeSafe's API. Liquid d1 (`liquid/d1`) and GPT-6 Luna Decisions (`openai/gpt-6-luna-decisions`) use Vercel AI Gateway, which does not take a model version. All three models get the same requests.
- Jev and GPT-6 Luna Decisions return each probability with 2 decimals. Liquid d1 returns full floating-point values.
- Every figure in a row uses the lists that the model answered. A refused list has no answer. Jev and Liquid d1 refused no lists.
- A table leaves out a column when no model has that figure. A list with no correct option has no accuracy. A list with only correct options has no distractor.
- The three differences and the probabilities are calculated for each answer, then averaged across answers. Differences involving a distractor use answers to lists with a correct option and a distractor. When a run also has lists with only correct options, the column "Lists with a correct option and a distractor" shows how many lists have both.
- The MMLU runs use a seeded sample of 1,000 of the 13,096 questions whose choices can be moved. The HotpotQA runs with 5 cyclic permutations and the HotpotQA run with 4 distractors and NOTA use a seeded sample of 1,000 of the 7,345 questions with 10 passages. The other runs use every example.
- MetaTool repeats queries across its test files. The run with paraphrases has 1,767 examples and 1,047 different requests. The run with duplicates has 3,292 examples and 3,275 different requests. An answer to a repeated request counts once per example.
- ECtHR labels name the articles that the Court found violated in a case. A case with no label means the Court found no violation of the 10 articles.
- The test sends a yes/no question about each tool of the 497 MetaTool multi-tool queries, run on 2026-10-14. It sends a yes/no question about each HotpotQA passage for 500 questions, and about each ECtHR article for the 1,000 test cases.
- The file `src/grjev/stimuli/tickets.json` has the tickets, TypeSafe's example request, and its documented answer. `python scripts/run/typesafe_example.py` sends that request.

Tables by run

1. Order: lists whose options are all correct

MetaTool: 5 duplicates of one correct tool

The list has 5 copies of one correct tool from the example. Their names differ only by extra spaces. Each example is sent with 5 cyclic permutations of the list. Each option appears at each position once. 3,292 examples and 16,460 lists. Each list has 5 options. The number of correct options in a list is 5.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	16,460	0	0.32	0.09	+0.21	0.321, 0.232, 0.186, 0.148, 0.112	2.21	0.23, 0.12, 0.17, 0.23, 0.25	5,956, 86, 494, 4,073, 5,851
Liquid d1	16,460	0	0.81	0.73	+0.78	0.806, 0.078, 0.054, 0.035, 0.026	1.03	0.81, 0.07, 0.03, 0.03, 0.06	16,445, 10, 0, 0, 5
GPT-6 Luna Decisions	16,198	262	0.78	0.66	+0.77	0.783, 0.127, 0.049, 0.025, 0.015	1.02	0.78, 0.10, 0.04, 0.03, 0.05	15,970, 5, 11, 43, 169

`results/metatool_copied/2026-10-11_01`, every list

MMLU: 5 duplicates of the correct choice

The list has 5 copies of the question's correct choice. Each example is sent with 5 cyclic permutations of the list. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 5.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.33	0.09	+0.23	0.334, 0.240, 0.188, 0.136, 0.102	2.18	0.14, 0.12, 0.19, 0.24, 0.31	296, 27, 160, 824, 3,693
Liquid d1	5,000	0	0.50	0.31	+0.42	0.502, 0.187, 0.133, 0.102, 0.077	1.85	0.48, 0.16, 0.13, 0.10, 0.13	4,283, 329, 219, 52, 117
GPT-6 Luna Decisions	4,832	168	0.74	0.58	+0.73	0.743, 0.164, 0.053, 0.026, 0.014	1.03	0.71, 0.17, 0.06, 0.03, 0.03	4,242, 379, 116, 55, 40

`results/mmlu_copied/2026-10-11_01`, every list

HotpotQA: 5 duplicates of one supporting passage

The list has 5 copies of one passage that supports the question. Each example is sent with 5 cyclic permutations of the list. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 5.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.34	0.11	+0.23	0.341, 0.226, 0.176, 0.143, 0.114	2.18	0.20, 0.18, 0.21, 0.19, 0.23	1,146, 664, 955, 761, 1,474
Liquid d1	5,000	0	0.73	0.63	+0.68	0.728, 0.101, 0.072, 0.057, 0.043	1.33	0.73, 0.09, 0.06, 0.05, 0.08	4,975, 10, 10, 0, 5
GPT-6 Luna Decisions	4,621	379	0.79	0.68	+0.78	0.792, 0.115, 0.049, 0.029, 0.015	1.00	0.79, 0.11, 0.04, 0.02, 0.05	4,595, 20, 3, 1, 2

`results/hotpotqa_copied/2026-10-11_01`, every list

ECtHR: 5 duplicates of one violated article

The list has 5 copies of one article that the Court found violated in the case. Each example is sent with 5 cyclic permutations of the list. Each option appears at each position once. 847 examples and 4,235 lists. Each list has 5 options. The number of correct options in a list is 5.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	4,235	0	0.29	0.06	+0.17	0.295, 0.236, 0.194, 0.154, 0.122	2.24	0.16, 0.13, 0.21, 0.23, 0.27	153, 11, 357, 1,203, 2,511
Liquid d1	4,235	0	0.55	0.36	+0.48	0.547, 0.189, 0.109, 0.084, 0.070	1.80	0.55, 0.19, 0.10, 0.08, 0.08	4,217, 18, 0, 0, 0
GPT-6 Luna Decisions	4,179	56	0.67	0.45	+0.65	0.665, 0.212, 0.069, 0.033, 0.020	1.35	0.66, 0.21, 0.07, 0.03, 0.03	4,134, 41, 0, 4, 0

`results/ecthr_copied/2026-10-11_01`, every list

MetaTool: 5 paraphrases of one correct tool

The list has 5 descriptions of one correct tool in 5 styles. Claude Opus 5.5 wrote them. Each example is sent with 5 cyclic permutations of the list. Each option appears at each position once. 1,767 examples and 8,835 lists. Each list has 5 options. The number of correct options in a list is 5.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	8,835	0	0.47	0.23	+0.42	0.468, 0.242, 0.152, 0.090, 0.047	1.85	0.23, 0.18, 0.17, 0.20, 0.21	2,336, 1,518, 1,279, 1,713, 1,989
Liquid d1	8,835	0	0.52	0.30	+0.46	0.517, 0.214, 0.124, 0.085, 0.059	1.80	0.45, 0.16, 0.12, 0.11, 0.17	6,571, 454, 415, 408, 987
GPT-6 Luna Decisions	8,155	680	0.59	0.37	+0.56	0.590, 0.223, 0.102, 0.054, 0.031	1.56	0.56, 0.19, 0.09, 0.06, 0.09	7,124, 670, 164, 81, 116

`results/metatool_rotated/2026-10-11_01`, every list

MetaTool: one correct tool at the first place of the list

The example's correct tool is first in its tool list. "None" is the last option. The test files are `similar_tools` and `scenario`. 2,795 examples and 2,795 lists. Each list has 6 to 16 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.78	0.87	+0.08	0.08	0.77	+0.53	0.744	0.216	0.48
Liquid d1	2,795	0	0.69	0.80	+0.10	0.11	0.65	+0.33	0.629	0.303	0.78
GPT-6 Luna Decisions	2,793	2	0.80	0.89	+0.09	0.09	0.80	+0.55	0.760	0.206	0.44

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `first`

MetaTool: one correct tool at the middle of the list

The example's correct tool is in the middle of its tool list. "None" is the last option. The test files are `similar_tools` and `scenario`. 2,795 examples and 2,795 lists. Each list has 6 to 16 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.80	0.88	+0.08	0.08	0.79	+0.57	0.765	0.200	0.42
Liquid d1	2,795	0	0.74	0.82	+0.08	0.09	0.69	+0.43	0.686	0.252	0.70
GPT-6 Luna Decisions	2,793	2	0.72	0.87	+0.15	0.15	0.76	+0.42	0.686	0.266	0.53

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `middle`

MetaTool: one correct tool at the last place of the list

The example's correct tool is last in its tool list. "None" is the last option. The test files are `similar_tools` and `scenario`. 2,795 examples and 2,795 lists. Each list has 6 to 16 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.80	0.88	+0.08	0.08	0.79	+0.56	0.763	0.202	0.43
Liquid d1	2,795	0	0.74	0.83	+0.09	0.09	0.70	+0.45	0.696	0.245	0.66
GPT-6 Luna Decisions	2,793	2	0.67	0.86	+0.19	0.19	0.75	+0.35	0.647	0.301	0.53

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `last`

2. Allocation: lists with several correct options

MetaTool: two correct tools, "Choose one of them."

The example's tool list has two correct tools. We send it in 9 orders: the released order, 5 orders with the two correct tools next to each other, and 3 orders with them apart. 497 examples and 4,473 lists. Each list has 10 options. The number of correct options in a list is 2.

Instruction: "Two tools in the list are appropriate to solve the user's query. Choose one of them."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	4,473	0	0.97	0.74	0.82	-0.15	0.15	0.67	+0.68	+0.77	+0.09	0.817, 0.136	21%	0.045	0.61
Liquid d1	4,473	0	0.95	0.72	0.77	-0.17	0.17	0.58	+0.60	+0.68	+0.08	0.758, 0.159	27%	0.075	0.77
GPT-6 Luna Decisions	4,473	0	0.93	0.60	0.79	-0.14	0.14	0.64	+0.64	+0.66	+0.02	0.761, 0.118	18%	0.097	0.81

`results/metatool_position/2026-10-11_02`, every list

MetaTool: two correct tools, "Choose both."

The example's tool list has two correct tools. We send it in 9 orders: the released order, 5 orders with the two correct tools next to each other, and 3 orders with them apart. 497 examples and 4,473 lists. Each list has 10 options. The number of correct options in a list is 2.

Instruction: "Two tools in the list are appropriate to solve the user's query. Choose both."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	4,473	0	0.98	0.82	0.82	-0.16	0.16	0.65	+0.66	+0.77	+0.11	0.809, 0.149	21%	0.039	0.65
Liquid d1	4,473	0	0.95	0.74	0.75	-0.20	0.20	0.54	+0.56	+0.65	+0.09	0.737, 0.173	30%	0.083	0.84
GPT-6 Luna Decisions	4,473	0	0.90	0.57	0.74	-0.16	0.16	0.58	+0.58	+0.57	-0.01	0.697, 0.116	16%	0.123	1.15

`results/metatool_position/2026-10-11_03`, every list

MetaTool: two correct tools, every needed tool is asked for

We use the example's tool list in the released order. 497 examples and 497 lists. Each list has 10 options. The number of correct options in a list is 2.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	497	0	0.98	0.73	0.86	-0.12	0.12	0.73	+0.74	+0.81	+0.07	0.847, 0.109	15%	0.042	0.52
Liquid d1	497	0	0.91	0.70	0.74	-0.16	0.17	0.53	+0.55	+0.59	+0.05	0.710, 0.162	27%	0.117	0.88
GPT-6 Luna Decisions	497	0	0.82	0.52	0.74	-0.08	0.08	0.56	+0.55	+0.49	-0.06	0.663, 0.110	16%	0.173	1.09

`results/metatool_removal/2026-10-11_01`, list `full`

HotpotQA: the passages of the question, "supports"

We use the question's passages in the released order. Two passages have a supporting fact. 7,405 examples and 7,405 lists. Each list has 2 to 10 options. The number of correct options in a list is 2.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.82	0.78	-0.19	0.19	0.60	+0.60	+0.72	+0.13	0.769, 0.172	30%	0.046	0.74
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.743, 0.193	35%	0.050	0.83
GPT-6 Luna Decisions	7,396	9	7,375	0.95	0.70	0.75	-0.20	0.20	0.57	+0.57	+0.65	+0.08	0.725, 0.154	26%	0.077	1.00

`results/hotpotqa_removal/2026-10-11_01`, list `full`

HotpotQA: the passages of the question, "may support"

We use the question's passages in the released order. Two passages have a supporting fact. 7,405 examples and 7,405 lists. Each list has 2 to 10 options. The number of correct options in a list is 2.

Instruction: "Choose every passage in the list that may support the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.82	0.78	-0.19	0.19	0.58	+0.59	+0.72	+0.13	0.765, 0.179	31%	0.044	0.75
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.742, 0.192	34%	0.051	0.85
GPT-6 Luna Decisions	7,401	4	7,380	0.94	0.70	0.75	-0.20	0.20	0.57	+0.57	+0.65	+0.08	0.725, 0.155	25%	0.078	0.99

`results/hotpotqa_removal/2026-10-11_01`, list `full_may`

HotpotQA: the passages of the question, "definitely supports"

We use the question's passages in the released order. Two passages have a supporting fact. 7,405 examples and 7,405 lists. Each list has 2 to 10 options. The number of correct options in a list is 2.

Instruction: "Choose every passage in the list that definitely supports the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.81	0.79	-0.18	0.18	0.60	+0.60	+0.73	+0.12	0.772, 0.169	29%	0.045	0.73
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.743, 0.192	34%	0.050	0.83
GPT-6 Luna Decisions	7,387	18	7,366	0.95	0.68	0.74	-0.20	0.20	0.57	+0.57	+0.64	+0.07	0.723, 0.151	25%	0.079	1.02

`results/hotpotqa_removal/2026-10-11_01`, list `full_definitely`

ECtHR: cases with 2 to 4 violated articles

The list has 10 articles. We send each example with the list's 10 cyclic permutations. Each option appears at each position once. 192 examples and 1,920 lists. Each list has 10 options. The number of correct options in a list is 2 to 4.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	1,920	0	0.95	0.76	0.74	-0.21	0.21	0.54	+0.58	+0.65	+0.08	highest 0.724, lowest 0.147	23%	0.069	0.89
Liquid d1	1,920	0	0.96	0.73	0.74	-0.23	0.23	0.54	+0.57	+0.66	+0.09	highest 0.725, lowest 0.155	25%	0.065	0.96
GPT-6 Luna Decisions	1,920	0	0.96	0.70	0.74	-0.21	0.21	0.57	+0.61	+0.67	+0.07	highest 0.732, lowest 0.125	17%	0.060	1.06

`results/ecthr_own/2026-10-11_01`, cases with 2 to 4 labels, every list

3. Abstention: lists with no correct option

MetaTool: 5 random tools, no "None"

The list has 5 tools chosen with a seed from the dataset. These tools are not in the example's tool list. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 3,292 examples and 16,460 lists. Each list has 5 options. The number of correct options in a list is 0.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	16,460	0	0.78	0.63	0.83	0.20, 0.21, 0.18, 0.20, 0.20	3,405, 3,457, 2,991, 3,328, 3,279
Liquid d1	16,460	0	0.74	0.60	0.98	0.19, 0.21, 0.21, 0.18, 0.22	3,260, 3,458, 3,383, 2,851, 3,508
GPT-6 Luna Decisions	6,950	9,510	0.79	0.67	0.94	0.25, 0.21, 0.19, 0.16, 0.19	1,568, 1,466, 1,313, 1,151, 1,452

`results/metatool_unrelated/2026-10-11_01`, every list

MMLU: 5 random choices, no NOTA

The list has 5 choices from other questions, chosen with a seed. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 0.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.55	0.34	1.66	0.22, 0.22, 0.17, 0.20, 0.20	1,083, 1,115, 768, 1,028, 1,006
Liquid d1	5,000	0	0.60	0.41	1.48	0.28, 0.24, 0.19, 0.12, 0.17	1,580, 1,258, 931, 467, 764
GPT-6 Luna Decisions	772	4,228	0.74	0.61	1.11	0.38, 0.23, 0.14, 0.11, 0.14	334, 179, 93, 78, 88

`results/mmlu_unrelated/2026-10-11_01`, every list

HotpotQA: 5 random passages, no NOTA

The list has 5 passages from other questions, chosen with a seed. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 0.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.58	0.38	1.55	0.21, 0.22, 0.18, 0.19, 0.20	1,012, 1,186, 881, 925, 996
Liquid d1	5,000	0	0.49	0.29	1.78	0.14, 0.25, 0.22, 0.16, 0.23	488, 1,473, 1,105, 637, 1,297
GPT-6 Luna Decisions	99	4,901	0.71	0.55	1.25	0.22, 0.18, 0.19, 0.09, 0.32	16, 19, 17, 10, 37

`results/hotpotqa_unrelated/2026-10-11_01`, every list

MetaTool: 4 distractors of the example, no "None"

The list has 4 tools from the example's tool list. None of the 4 tools is correct. The study sends the list in 4 cyclic permutations for each example. Each option appears at each position once. 995 examples and 3,980 lists. Each list has 4 options. The number of correct options in a list is 0.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	3,980	0	0.80	0.65	0.68	0.26, 0.26, 0.24, 0.25	1,036, 1,017, 939, 988
Liquid d1	3,980	0	0.77	0.61	0.82	0.27, 0.26, 0.25, 0.22	1,117, 1,023, 991, 849
GPT-6 Luna Decisions	3,059	921	0.78	0.63	0.86	0.29, 0.27, 0.24, 0.20	916, 826, 711, 606

`results/metatool_distractors/2026-10-11_01`, every list

MMLU: the 3 distractors of the question, no NOTA

The list has the question's 3 incorrect choices. The study sends the list in 3 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 3,000 lists. Each list has 3 options. The number of correct options in a list is 0.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 1.58)	Mean probability at each position	Predictions at each position
Jev	3,000	0	0.73	0.53	0.89	0.35, 0.34, 0.32	1,059, 1,018, 923
Liquid d1	3,000	0	0.76	0.58	0.82	0.33, 0.34, 0.33	1,013, 1,025, 962
GPT-6 Luna Decisions	2,832	168	0.77	0.59	0.80	0.42, 0.30, 0.29	1,246, 796, 790

`results/mmlu_distractors/2026-10-11_01`, every list

MetaTool: the tool list without its two correct tools

The list has the example's tools in their released order. It leaves out the two correct tools. 497 examples and 497 lists. Each list has 8 options. The number of correct options in a list is 0.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 3.00)
Jev	497	0	0.82	0.68	0.67
Liquid d1	497	0	0.79	0.67	0.85
GPT-6 Luna Decisions	405	92	0.76	0.62	1.06

`results/metatool_removal/2026-10-11_01`, list `without_all`

HotpotQA: the passages without the supporting passages

The list has the question's passages in their released order. It leaves out passages that hold a supporting fact. 7,377 examples and 7,377 lists. Each list has 2 to 8 options. The number of correct options in a list is 0.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 1.00 to 3.00)
Jev	7,377	0	0.61	0.43	1.51
Liquid d1	7,377	0	0.67	0.51	1.32
GPT-6 Luna Decisions	5,352	2,025	0.65	0.50	1.48

`results/hotpotqa_removal/2026-10-11_01`, list `without_all`

ECtHR: the articles without the violated articles

The list has the 10 articles in their released order. It leaves out the articles the Court found violated. 847 examples and 847 lists. Each list has 6 to 9 options. The number of correct options in a list is 0.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.58 to 3.17)
Jev	847	0	0.74	0.57	0.99
Liquid d1	847	0	0.74	0.60	1.09
GPT-6 Luna Decisions	610	237	0.71	0.58	1.39

`results/ecthr_removal/2026-10-11_01`, list `without_all`

ECtHR: cases with no violated article, no NOTA

The list has the 10 articles. The study sends the list in 10 cyclic permutations for each example. Each option appears at each position once. 153 examples and 1,530 lists. Each list has 10 options. The number of correct options in a list is 0.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 3.32)
Jev	1,530	0	0.83	0.71	0.66
Liquid d1	1,530	0	0.87	0.77	0.54
GPT-6 Luna Decisions	1,515	15	0.87	0.78	0.62

`results/ecthr_own/2026-10-11_01`, cases with no label, every list

MetaTool: 4 random tools and "None"

The list has 4 tools from the dataset that are not in the example's tool list. It also has "None", the correct option. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 995 examples and 4,975 lists. Each list has 5 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	4,975	0	0.95	0.95	0.00	0.01	0.91	+0.86	0.929	0.068	0.16	0.20, 0.20, 0.20, 0.20, 0.20	990, 1,009, 972, 1,002, 1,002
Liquid d1	4,975	0	0.98	0.97	-0.01	0.01	0.93	+0.92	0.957	0.040	0.14	0.20, 0.20, 0.20, 0.20, 0.20	985, 995, 999, 990, 1,006
GPT-6 Luna Decisions	4,966	9	0.93	0.95	+0.02	0.02	0.91	+0.82	0.902	0.087	0.24	0.20, 0.20, 0.20, 0.20, 0.19	1,014, 999, 987, 987, 979

`results/metatool_absent_random/2026-10-11_01`, every list

MMLU: 4 random choices and NOTA

The list has 4 choices from other questions and NOTA, the correct option. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.97	0.95	-0.03	0.03	0.91	+0.89	0.936	0.049	0.26	0.20, 0.20, 0.20, 0.20, 0.20	1,006, 998, 987, 1,001, 1,008
Liquid d1	5,000	0	0.99	0.98	-0.01	0.01	0.96	+0.95	0.973	0.021	0.10	0.20, 0.20, 0.20, 0.20, 0.20	1,005, 997, 1,001, 998, 999
GPT-6 Luna Decisions	4,996	4	0.99	0.94	-0.05	0.05	0.90	+0.90	0.936	0.041	0.34	0.21, 0.21, 0.20, 0.19, 0.19	1,007, 1,002, 996, 995, 996

`results/mmlu_absent_random/2026-10-11_01`, every list

HotpotQA: 4 random passages and NOTA

The list has 4 passages from other questions and NOTA, the correct option. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	1.00	1.00	0.00	0.00	0.99	+0.99	0.995	0.004	0.03	0.20, 0.20, 0.20, 0.20, 0.20	1,000, 1,000, 1,000, 1,000, 1,000
Liquid d1	5,000	0	1.00	1.00	0.00	0.00	1.00	+1.00	1.000	0.000	0.00	0.20, 0.20, 0.20, 0.20, 0.20	1,000, 1,000, 1,000, 1,000, 1,000
GPT-6 Luna Decisions	5,000	0	1.00	0.94	-0.06	0.06	0.91	+0.91	0.943	0.036	0.34	0.22, 0.19, 0.20, 0.20, 0.20	1,001, 1,000, 1,000, 999, 1,000

`results/hotpotqa_absent_random/2026-10-11_01`, every list

MMLU: the 3 distractors of the question and NOTA

The list has the question's 3 incorrect choices and NOTA, the correct option. The study sends the list in 4 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 4,000 lists. Each list has 4 options. The number of correct options in a list is 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	4,000	0	0.65	0.77	+0.12	0.12	0.61	+0.23	0.575	0.341	0.85	0.26, 0.25, 0.24, 0.25	1,053, 998, 971, 978
Liquid d1	4,000	0	0.72	0.81	+0.10	0.10	0.68	+0.40	0.668	0.273	0.69	0.22, 0.25, 0.27, 0.26	880, 978, 1,097, 1,045
GPT-6 Luna Decisions	4,000	0	0.54	0.76	+0.22	0.22	0.59	+0.10	0.497	0.395	0.89	0.28, 0.25, 0.25, 0.23	1,183, 963, 940, 914

`results/mmlu_absent/2026-10-11_01`, every list

HotpotQA: 4 distractors of the question and NOTA

The list has 4 passages from the question that hold no supporting fact. It also has NOTA, the correct option. The study sends the list in 5 cyclic permutations for each example. Each option appears at each position once. 1,000 examples and 5,000 lists. Each list has 5 options. The number of correct options in a list is 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.84	0.82	-0.01	0.03	0.69	+0.55	0.752	0.199	0.67	0.19, 0.21, 0.19, 0.20, 0.21	973, 1,047, 970, 998, 1,012
Liquid d1	5,000	0	0.88	0.90	+0.02	0.02	0.82	+0.70	0.840	0.136	0.39	0.19, 0.20, 0.20, 0.20, 0.21	975, 999, 1,015, 992, 1,019
GPT-6 Luna Decisions	5,000	0	0.79	0.77	-0.01	0.04	0.63	+0.44	0.677	0.232	0.92	0.24, 0.20, 0.20, 0.18, 0.18	1,184, 967, 998, 905, 946

`results/hotpotqa_absent/2026-10-11_01`, every list

MetaTool `reliability`: the tool list of the example and "None"

The example's tools stay in their released order, with "None" as the last option. No tool in the list is correct. "None" is the correct option. 995 examples and 995 lists. Each list has 11 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	995	0	0.91	0.91	+0.01	0.02	0.84	+0.74	0.864	0.127	0.31
Liquid d1	995	0	0.98	0.94	-0.03	0.03	0.90	+0.88	0.933	0.056	0.25
GPT-6 Luna Decisions	993	2	0.94	0.95	+0.01	0.03	0.92	+0.84	0.912	0.077	0.21

`results/metatool_position/2026-10-11_01`, test file `reliability`, list `released`

ECtHR: the 10 articles and NOTA, cases with no violated article

The list has the 10 articles in their released order, with NOTA as the last option. NOTA is the correct option. 153 examples and 153 lists. Each list has 11 options. The number of correct options in a list is 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	153	0	0.37	0.66	+0.29	0.30	0.39	-0.13	0.392	0.524	1.09
Liquid d1	153	0	0.07	0.79	+0.72	0.72	0.64	-0.64	0.132	0.767	0.81
GPT-6 Luna Decisions	153	0	0.17	0.72	+0.55	0.55	0.54	-0.42	0.233	0.652	1.09

`results/ecthr_with_none/2026-10-11_01`, cases with no label, list `own`

4. Calibration: lists with one correct option

MetaTool: one correct tool, the 6 orders of the list

The run uses the example's tool list in its released order. It also puts the correct tool in 5 places. "None" is the last option. The test files are `similar_tools` and `scenario`. 2,795 examples and 16,770 lists. Each list has 6 to 16 options. The number of correct options in a list is 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	16,770	0	0.79	0.88	+0.08	0.08	0.78	+0.55	0.756	0.207	0.45
Liquid d1	16,770	0	0.72	0.81	+0.09	0.09	0.67	+0.40	0.669	0.268	0.72
GPT-6 Luna Decisions	16,758	12	0.73	0.87	+0.14	0.14	0.77	+0.45	0.701	0.256	0.49

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, every list

MMLU: the 4 choices of the question

The model sees the question's 4 choices. Each example uses 4 cyclic permutations, so every option appears at each position once. 1,000 examples and 4,000 lists. Each list has 4 options. The number of correct options in a list is 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	4,000	0	0.92	0.94	+0.02	0.04	0.89	+0.77	0.878	0.104	0.26	0.25, 0.25, 0.25, 0.25	1,000, 1,001, 995, 1,004
Liquid d1	4,000	0	0.84	0.89	+0.05	0.05	0.81	+0.65	0.807	0.153	0.42	0.24, 0.25, 0.25, 0.25	971, 1,011, 1,022, 996
GPT-6 Luna Decisions	3,989	11	0.83	0.88	+0.04	0.05	0.80	+0.64	0.796	0.157	0.44	0.26, 0.25, 0.25, 0.24	1,060, 989, 998, 942

`results/mmlu_own/2026-10-11_01`, every list

ECtHR: cases with 1 violated article

The model sees the 10 articles. Each example uses 10 cyclic permutations, so every option appears at each position once. 655 examples and 6,550 lists. Each list has 10 options. The number of correct options in a list is 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	6,550	0	0.89	0.86	-0.03	0.03	0.75	+0.66	0.818	0.156	0.52
Liquid d1	6,550	0	0.88	0.87	-0.01	0.01	0.77	+0.66	0.819	0.155	0.51
GPT-6 Luna Decisions	6,550	0	0.90	0.89	0.00	0.01	0.82	+0.72	0.845	0.121	0.47

`results/ecthr_own/2026-10-11_01`, cases with 1 label, every list

ECtHR: the 10 articles and NOTA, cases with 1 violated article

The model sees the 10 articles in the released order, with NOTA as the last option. 655 examples and 655 lists. Each list has 11 options. The number of correct options in a list is 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	655	0	0.84	0.74	-0.10	0.11	0.54	+0.44	0.682	0.242	0.95
Liquid d1	655	0	0.88	0.86	-0.03	0.04	0.76	+0.65	0.800	0.154	0.62
GPT-6 Luna Decisions	655	0	0.87	0.83	-0.05	0.05	0.72	+0.63	0.778	0.153	0.76

`results/ecthr_with_none/2026-10-11_01`, cases with 1 label, list `own`

5. Dependence on the list: a correct option is removed

MetaTool: the tool list without one of its two correct tools

The tools appear in the released order, with one correct tool removed. Each example gives 2 lists, one for each removed tool. 497 examples and 994 lists. Each list has 9 options. The number of correct options in a list is 1.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.17)
Jev	994	0	0.86	0.90	+0.04	0.04	0.81	+0.66	0.821	0.163	0.36
Liquid d1	994	0	0.78	0.85	+0.07	0.07	0.72	+0.50	0.732	0.237	0.57
GPT-6 Luna Decisions	987	7	0.67	0.82	+0.15	0.15	0.70	+0.32	0.620	0.303	0.80

`results/metatool_removal/2026-10-11_01`, lists `without_1` and `without_2`

HotpotQA: the passages without one supporting passage

The passages appear in the released order, with one supporting passage removed. Each question gives one list for each removed passage. 7,384 examples and 14,768 lists. Each list has 2 to 9 options. The number of correct options in a list is 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.17)
Jev	14,768	0	0.84	0.83	-0.02	0.03	0.72	+0.59	0.758	0.169	0.72
Liquid d1	14,768	0	0.84	0.85	+0.01	0.02	0.76	+0.61	0.778	0.165	0.61
GPT-6 Luna Decisions	14,327	441	0.80	0.80	0.00	0.03	0.69	+0.51	0.703	0.194	0.92

`results/hotpotqa_removal/2026-10-11_01`, lists `without_1` and `without_2`

ECtHR: the articles without one violated article

The 10 articles appear in the released order, with one violated article removed. This applies to cases with 2 to 4 violated articles. Each case gives one list for each removed article. 192 examples and 435 lists. Each list has 9 options. The number of correct options in a list is 1 to 3.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.17)
Jev	435	0	0.90	0.86	0.83	-0.07	0.07	0.70	+0.19	+0.66	+0.47	highest 0.785, lowest 0.593	68%	0.126	0.64
Liquid d1	435	0	0.90	0.81	0.83	-0.07	0.07	0.71	+0.19	+0.65	+0.46	highest 0.779, lowest 0.585	68%	0.130	0.70
GPT-6 Luna Decisions	426	9	0.92	0.85	0.84	-0.08	0.08	0.75	+0.23	+0.71	+0.48	highest 0.805, lowest 0.576	65%	0.096	0.75

`results/ecthr_removal/2026-10-11_01`, lists `without_1` to `without_4`

MetaTool: the second correct option before and after the removal

An example's second correct option is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability	Examples in which the options other than the removed one hold 0.1 or more	On these examples: its mean probability in the full list, renormalised over those options	On these examples: its mean probability in the list without the removed option
Jev	497	0.11	0.71	227	0.77	0.72
Liquid d1	497	0.16	0.62	388	0.64	0.61
GPT-6 Luna Decisions	490	0.11	0.53	385	0.40	0.49

`results/metatool_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

HotpotQA: the second correct option before and after the removal

An example's second correct option is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability	Examples in which the options other than the removed one hold 0.1 or more	On these examples: its mean probability in the full list, renormalised over those options	On these examples: its mean probability in the list without the removed option
Jev	7,384	0.17	0.66	4,866	0.80	0.67
Liquid d1	7,384	0.19	0.68	5,513	0.80	0.70
GPT-6 Luna Decisions	7,044	0.16	0.60	5,041	0.61	0.60

`results/hotpotqa_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

ECtHR: the second correct option before and after the removal

An example's second correct option is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability	Examples in which the options other than the removed one hold 0.1 or more	On these examples: its mean probability in the full list, renormalised over those options	On these examples: its mean probability in the list without the removed option
Jev	192	0.19	0.71	150	0.72	0.71
Liquid d1	192	0.16	0.68	138	0.70	0.67
GPT-6 Luna Decisions	183	0.15	0.68	143	0.59	0.68

`results/ecthr_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

MetaTool: a yes/no question per tool

We ask a separate yes/no question about each tool of the 497 multi-tool queries. We send the 10 questions for a query in one request.

Instruction: "Is this tool needed to solve the user's query? Tool: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	4,970	0	0.64	727 of 994	0.09	271 of 3,976
Liquid d1	4,970	0	0.79	841 of 994	0.09	289 of 3,976
GPT-6 Luna Decisions	4,970	0	0.65	660 of 994	0.06	222 of 3,976

`results/metatool_yes_no/2026-10-14_01`, every question

HotpotQA: a yes/no question per passage

We ask a separate yes/no question about each passage. We send the 10 questions for an example in one request.

Instruction: "Does this passage support the answer to the question? Passage: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	4,955	0	0.45	396 of 1,000	0.09	131 of 3,955
Liquid d1	4,955	0	0.40	375 of 1,000	0.06	125 of 3,955
GPT-6 Luna Decisions	4,951	4	0.52	508 of 1,000	0.06	181 of 3,951

`results/hotpotqa_yes_no/2026-10-11_01`, every question

HotpotQA: a yes/no question per passage, bridge questions

The yes/no run on HotpotQA, on its bridge questions, which need a fact from one passage to find the other. HotpotQA gives the type of each question.

Instruction: "Does this passage support the answer to the question? Passage: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	3,989	0	0.48	377 of 802	0.09	121 of 3,187
Liquid d1	3,989	0	0.45	352 of 802	0.06	111 of 3,187
GPT-6 Luna Decisions	3,989	0	0.57	463 of 802	0.06	168 of 3,187

`results/hotpotqa_yes_no/2026-10-11_01`, every question of the HotpotQA type `bridge`

HotpotQA: a yes/no question per passage, comparison questions

The yes/no run on HotpotQA, on its comparison questions, which compare two entities, one per passage. HotpotQA gives the type of each question.

Instruction: "Does this passage support the answer to the question? Passage: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	966	0	0.29	19 of 198	0.09	10 of 768
Liquid d1	966	0	0.22	23 of 198	0.04	14 of 768
GPT-6 Luna Decisions	962	4	0.30	45 of 198	0.06	13 of 764

`results/hotpotqa_yes_no/2026-10-11_01`, every question of the HotpotQA type `comparison`

ECtHR: a yes/no question per article

We ask a separate yes/no question about each of the 10 articles. We send the 10 questions for a case in one request.

Instruction: "Did the European Court of Human Rights find a violation of this article in this case? Article: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	10,000	0	0.67	964 of 1,090	0.15	607 of 8,910
Liquid d1	10,000	0	0.89	1,050 of 1,090	0.12	790 of 8,910
GPT-6 Luna Decisions	9,886	114	0.74	818 of 1,018	0.07	374 of 8,868

`results/ecthr_yes_no/2026-10-11_01`, every question

6. The documented example: support tickets for TypeSafe's team question

Tickets: two issues for two teams

Each ticket describes two problems, and each problem belongs to a different team. Each pair of problems is sent in both orders. The list has the 3 teams from TypeSafe's example in the same order: returns, shipping and billing. 24 examples and 24 lists. Each list has 3 options. The number of correct options in a list is 2.

Instruction: "Which team should handle this?"

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Answers with every correct option above 0.25	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.58)
Jev	24	0	1.00	1.00	0.76	-0.24	0.24	0.51	+0.51	+0.75	+0.24	0.755, 0.245	46%	0.000	0.72
Liquid d1	24	0	1.00	1.00	0.71	-0.29	0.29	0.44	+0.44	+0.71	+0.27	0.715, 0.279	50%	0.006	0.80
GPT-6 Luna Decisions	24	0	1.00	1.00	0.87	-0.13	0.13	0.75	+0.75	+0.87	+0.12	0.874, 0.122	12%	0.004	0.47

`results/tickets_removal/2026-10-11_01`, test file `two_issues`, list `full`

Tickets: one issue for one team

Each ticket describes one issue from the tickets above. The list has the 3 teams from TypeSafe's example in the same order: returns, shipping and billing. 24 examples and 24 lists. Each list has 3 options. The number of correct options in a list is 1.

Instruction: "Which team should handle this?"

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.58)
Jev	24	0	1.00	1.00	0.00	0.00	1.00	+1.00	1.000	0.000	0.00
Liquid d1	24	0	1.00	0.99	-0.01	0.01	0.99	+0.99	0.994	0.005	0.03
GPT-6 Luna Decisions	24	0	1.00	1.00	0.00	0.00	1.00	+1.00	0.998	0.001	0.01

`results/tickets_removal/2026-10-11_01`, test file `one_issue`, list `full`

Tickets: two issues, the list without one of the two teams

Each ticket has two issues. Its list has 2 teams: one of the ticket's two teams and the third team. 24 examples and 48 lists. Each list has 2 options. The number of correct options in a list is 1.

Instruction: "Which team should handle this?"

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00)
Jev	48	0	1.00	0.99	-0.01	0.01	0.98	+0.98	0.989	0.011	0.06
Liquid d1	48	0	1.00	0.98	-0.02	0.02	0.96	+0.96	0.978	0.022	0.11
GPT-6 Luna Decisions	48	0	1.00	0.98	-0.02	0.02	0.95	+0.95	0.977	0.023	0.11

`results/tickets_removal/2026-10-11_01`, test file `two_issues`, lists `without_1` and `without_2`

Tickets: one issue, the list without its team

Each ticket has one issue. Its list has the 2 teams that the issue does not concern. 24 examples and 24 lists. Each list has 2 options. The number of correct options in a list is 0.

Instruction: "Which team should handle this?"

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 1.00)
Jev	24	0	0.95	0.89	0.22
Liquid d1	24	0	0.83	0.67	0.58
GPT-6 Luna Decisions	20	4	0.95	0.90	0.24

`results/tickets_removal/2026-10-11_01`, test file `one_issue`, list `without_all`

Tickets: the ticket of TypeSafe's example

The ticket in TypeSafe's documentation describes a late delivery, a wrong size and a double charge. The list has the 3 teams from TypeSafe's example in the same order: returns, shipping and billing. The three teams are labelled as correct. 1 examples and 1 lists. Each list has 3 options. The number of correct options in a list is 3.

Instruction: "Which team should handle this?"

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 1.58)
Jev	1	0	0.58	0.21	+0.53	0.580, 0.370, 0.050	1.20
Liquid d1	1	0	0.75	0.55	+0.70	0.750, 0.202, 0.048	0.99
GPT-6 Luna Decisions	1	0	0.50	0.01	+0.49	0.500, 0.490, 0.010	1.07

`results/tickets_removal/2026-10-11_01`, test file `typesafe_example`, list `full`